

Using wearable data to predict speed and injury risk in runners

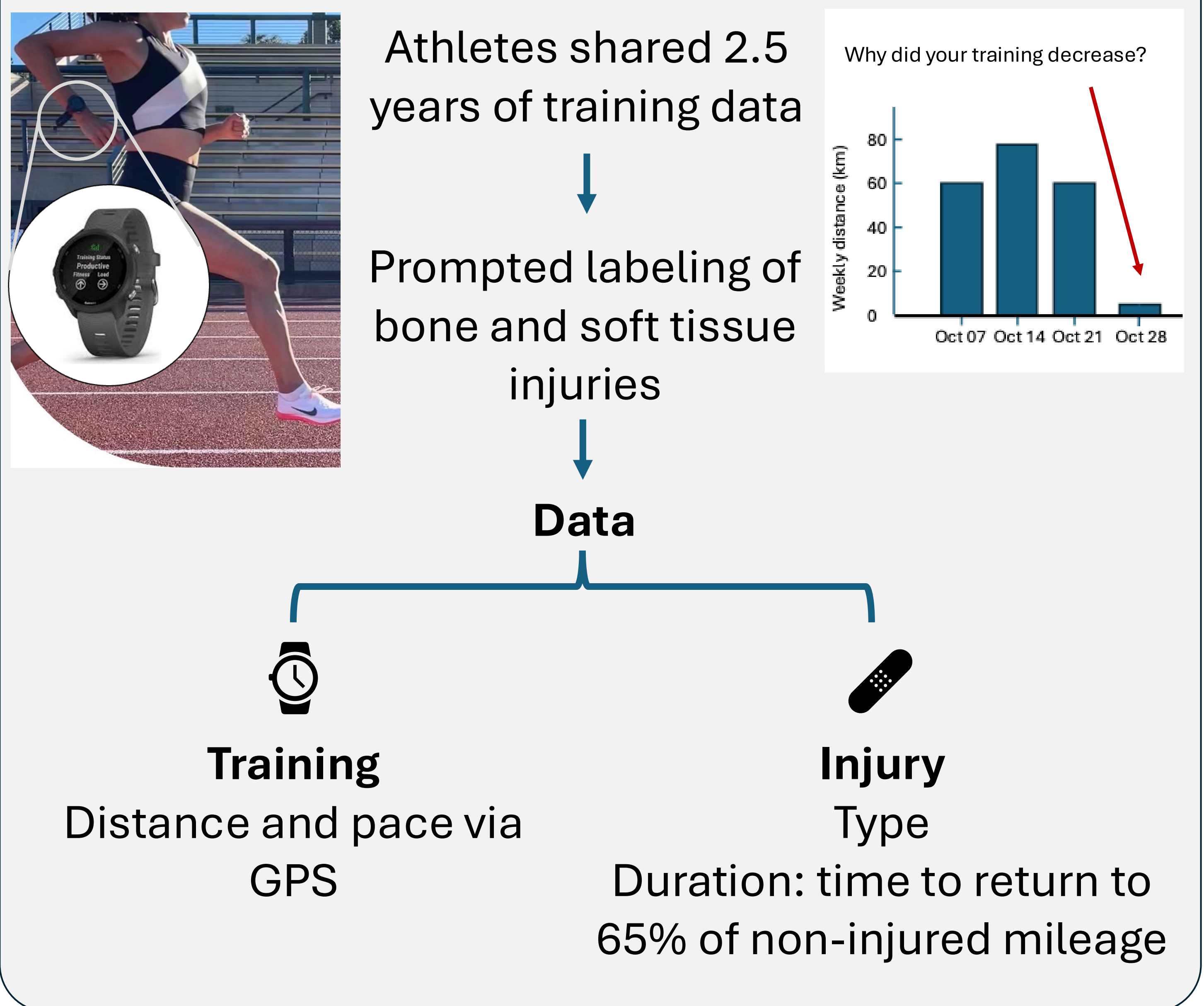
Sarah Johnson, Alex Gonzalez, Johanna O’Day, Ellen Kuhl, Scott Delp, Jennifer Hicks
Wu Tsai Human Performance Alliance, Stanford University

P1-256

Motivation

- GPS-enabled smartwatches provide large-scale physiological and biomechanical data that could help guide injury prevention for runners.
- We quantify the trade off between running performance and injury risk in a large, real-world cohort.

Methods



Cohort

	Male (n=56)	Female (n=157)
Age	31±6	30±6
BMI	22±2	21±2
Weekly run (km)	71±3	63±2
Soft tissue injury incidence*	0.49	0.68
Bone stress injury (BSI) incidence*	0.1	0.16
No. of soft tissue injuries	60	236
No. of BSIs	10	54

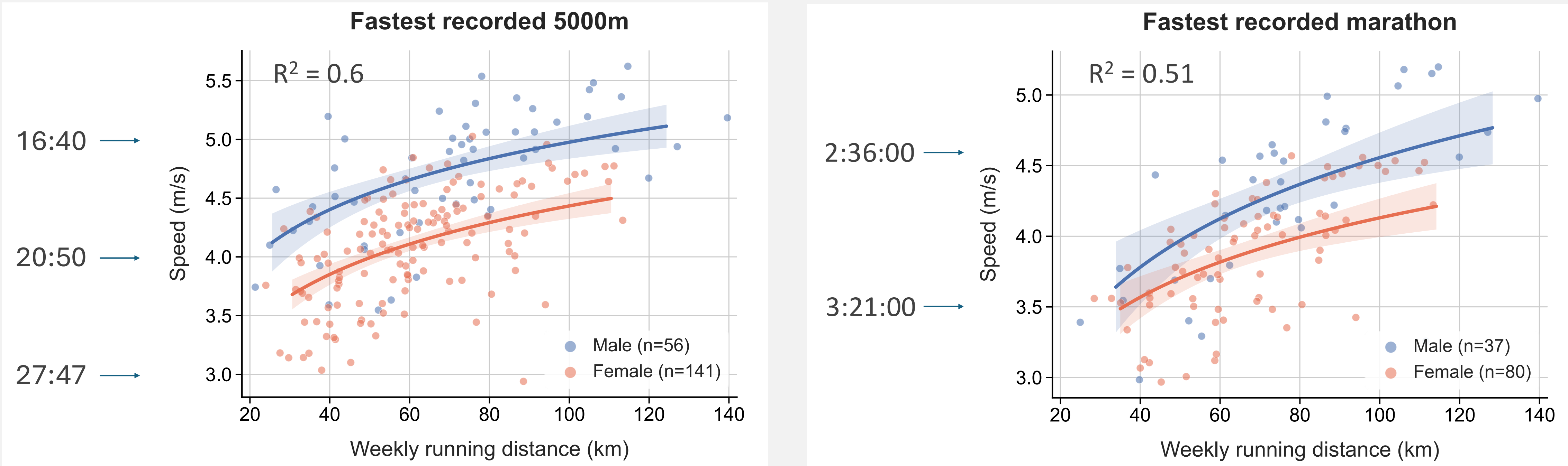
Injury Types

Achilles tendinitis 10%
Ankle sprain 2%
Hamstring 8%
Knee 16%
Lower leg muscle 8%
Lower leg tendon 8%
BSI 18%
Sciatica 2%
Plantar fasciitis 6%
Other 22%

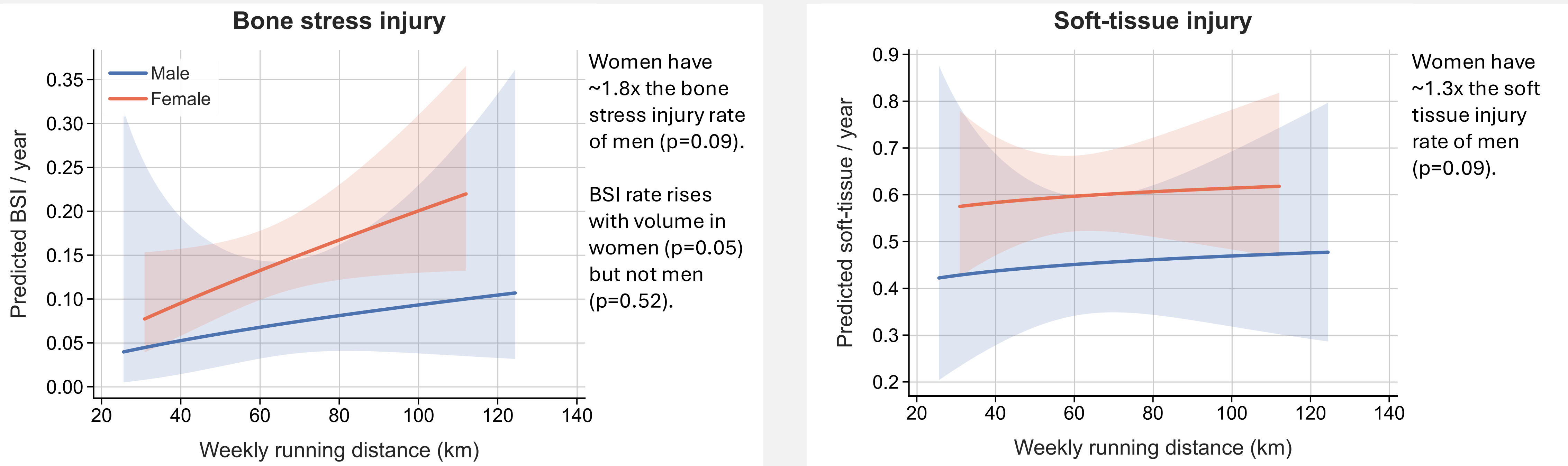
Results

Mean fastest recorded 5000m: 16:22 18:44

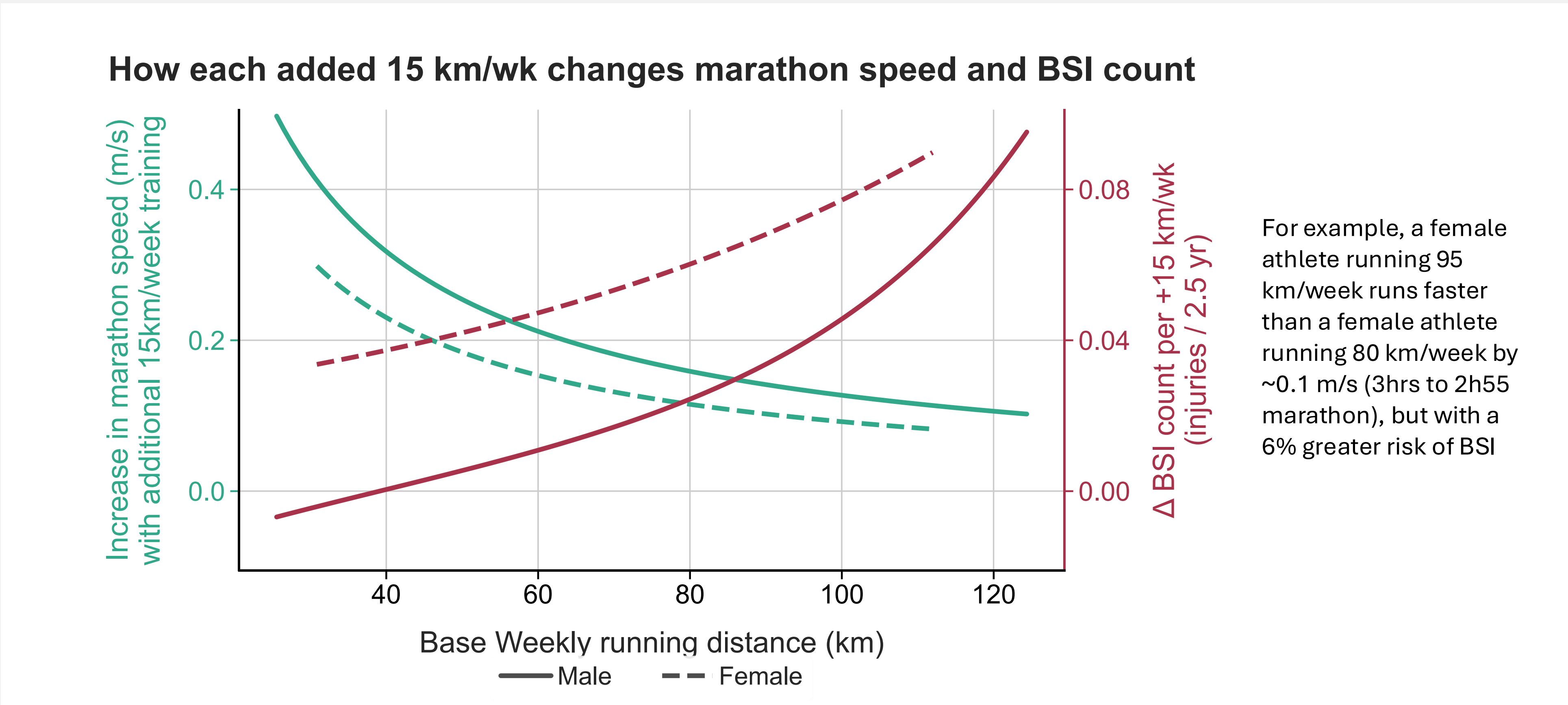
Training volume was associated with faster 5k and marathon speeds



Training volume was associated with bone stress injury, but not soft tissue injury rates



Higher mileage is associated with increased rates of BSI, along with improved performance, but diminishing returns



Wearables can quantify the interaction between training, injury risk, and performance with unprecedented granularity. Future work adding menstrual status may further inform training and injury risk.